

Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z2,ZL

Generated by MacAnalog-Symbolix

October 17, 2025

Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple **Z2 ZL:** $\frac{Z_2Z_Lg_m+Z_L}{Z_2g_m+1}$

$$H(z)=\frac{Z_2Z_Lg_m+Z_L}{Z_2g_m+1}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

8 X-INVALID-NUMER

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s)=(\infty, R_2, \infty, \infty, \infty, R_L)$

$$H(s)=R_L$$

9.2 X-INVALID-ORDER-2 $Z(s)=\left(\infty, R_2, \infty, \infty, \infty, \frac{1}{C_Ls}\right)$

$$H(s)=\frac{1}{C_Ls}$$

9.3 X-INVALID-ORDER-3 $Z(s)=\left(\infty, R_2, \infty, \infty, \infty, \frac{R_L}{C_LR_Ls+1}\right)$

$$H(s)=\frac{R_L}{C_LR_Ls+1}$$

9.4 X-INVALID-ORDER-4 $Z(s)=\left(\infty, R_2, \infty, \infty, \infty, R_L+\frac{1}{C_Ls}\right)$

$$H(s)=\frac{R_2g_m+s(C_LR_2R_Lg_m+C_LR_L)+1}{s(C_LR_2g_m+C_L)}$$

9.5 X-INVALID-ORDER-5 $Z(s)=\left(\infty, \frac{1}{C_2s}, \infty, \infty, \infty, R_L\right)$

$$H(s)=R_L$$

$$\mathbf{9.6 \quad X-INVALID-ORDER-6} \quad Z(s) = \left(\infty, \frac{1}{C_2 s}, \infty, \infty, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{1}{C_L s}$$

$$\mathbf{9.7 \quad X-INVALID-ORDER-7} \quad Z(s) = \left(\infty, \frac{1}{C_2 s}, \infty, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_L}{C_L R_L s + 1}$$

$$\mathbf{9.8 \quad X-INVALID-ORDER-8} \quad Z(s) = \left(\infty, \frac{1}{C_2 s}, \infty, \infty, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_2 C_L R_L s^2 + g_m + s (C_2 + C_L R_L g_m)}{C_2 C_L s^2 + C_L g_m s}$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = \left(\infty, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, R_L \right)$$

$$H(s) = R_L$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(\infty, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{1}{C_L s}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(\infty, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_L}{C_L R_L s + 1}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(\infty, \frac{R_2}{C_2 R_2 s + 1}, \infty, \infty, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_2 C_L R_2 R_L s^2 + R_2 g_m + s (C_2 R_2 + C_L R_2 R_L g_m + C_L R_L) + 1}{C_2 C_L R_2 s^2 + s (C_L R_2 g_m + C_L)}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(\infty, R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, R_L \right)$$

$$H(s) = R_L$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(\infty, R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{1}{C_L s}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(\infty, R_2 + \frac{1}{C_2 s}, \infty, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_L}{C_L R_L s + 1}$$

9.16 **X-INVALID-ORDER-16** $Z(s) = \left(\infty, \ R_2 + \frac{1}{C_2 s}, \ \infty, \ \infty, \ \infty, \ R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m + s^2 \left(C_2 C_L R_2 R_L g_m + C_2 C_L R_L \right) + s \left(C_2 R_2 g_m + C_2 + C_L R_L g_m \right)}{C_L g_m s + s^2 \left(C_2 C_L R_2 g_m + C_2 C_L \right)}$$

10 **X-INVALID-WZ**

11 **X-PolynomialError**